
D-GAMM: A Multi-Turn Benchmark for Dark-Patterns and Gradual Autonomy Manipulation¹

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Abstract

Most benchmarks for AI manipulation evaluate single model outputs and therefore miss risks that emerge across interaction over time. We introduce D-GAMM, a lightweight multi-turn probe designed to detect gradual autonomy interference and psychological destabilisation in large language model (LLM) conversations. Manipulation is operationalised as interactional behavior that escalates pressure after resistance, narrows perceived options, or reinforces destabilising interpretations. Using six short scenarios tested in baseline and vulnerable variants, we manually evaluate several state-of-the-art conversational models. Results show that manipulative dynamics can surface across turns, even when individual responses appear benign, revealing limitations of single-turn evaluations. We also observe that personalization and contextual inference may modulate these dynamics, raising additional governance concerns.

Keywords: Multiturn, autonomy interference, dark patterns

1. Introduction

Large language models are increasingly deployed in conversational settings where users seek advice, reassurance, or guidance on consequential decisions. In these contexts, safety risks do not arise only from isolated harmful outputs, but from interactional dynamics that shape how users reason, decide, and interpret their situation over time.

¹ Research conducted at the [AI Manipulation Hackathon](#), 2026

Prior work has documented behaviors such as sycophancy, strategic deception, reward hacking, and dark patterns, including benchmarks like DarkBench² that probe manipulative output patterns in single-turn settings. However, most existing evaluations assess manipulation at the level of isolated prompt–response pairs. This creates a mismatch with real-world interactions, where pressure, agreement, or framing may escalate gradually across turns, particularly following user hesitation or implicit vulnerability.

To address this gap, we propose a minimal, structured multi-turn evaluation probe. Rather than simulating realistic users or measuring persuasion success, the probe focuses on whether an LLM interferes with decisional autonomy or mental integrity across interaction. The aim is not to introduce a comprehensive benchmark, but to demonstrate a tractable, monitoring-oriented method for detecting manipulation that single-turn evaluations overlook.

We have chosen to work on those research questions: Can manipulation-related behaviors emerge across multiple interaction turns even when individual responses appear neutral or acceptable? And, do implicit vulnerability cues in user language correlate with increased severity or escalation of autonomy interference?

The contribution of this work is twofold. First, we introduce a process-based operationalisation of manipulation as gradual interference with autonomy and mental integrity, rather than intent or persuasion success. Second, we present a lightweight evaluation design suitable for post-deployment monitoring and governance contexts. D-GAMM can be understood as a process-oriented extension to output-focused benchmarks such as DarkBench, targeting interactional escalation rather than static response content.

2. Methods

We assume a threat model in which an LLM does not explicitly intend to manipulate, but nevertheless learns interaction strategies that influence users over time. These strategies may include escalating pressure after resistance, excessive validation, or reinforcing destabilising interpretations. The resulting risk is gradual erosion of user autonomy, particularly for vulnerable users, which may not be detected by single-turn safety evaluations.

Scenario Design

To assess model safety, we conducted a multi-turn experiment across six scenarios (N=6) covering domains where autonomy interference is plausibly harmful (decision-making under pressure, workplace power asymmetry, psychological distress, suspicion and belief formation, meaning-seeking and agreement/sycophancy edge cases). Each scenario was presented in two variants: Baseline (using neutral user phrasing) and Vulnerable (where vulnerability is expressed implicitly through user language (e.g. stress, isolation, fear of consequences)). Responses were annotated across four dimensions: Autonomy

² <https://github.com/smarter/darkbench>

Interference (0,1), Severity (0–3), Escalation (0–3), and Psychological Destabilization (0,1).

Baseline and vulnerable variants differ only in the initial user turn, ensuring that later behavioral differences reflect sensitivity to vulnerability rather than changes in resistance phrasing. No personas, system prompts, or role instructions are provided; interactions are conducted as natural user–model conversations. Scenarios were manually constructed for this study, informed by prior work on manipulation, sycophancy, and persuasion, but designed specifically to probe multi-turn escalation and autonomy interference. Manual testing was chosen over automated simulation to approximate real user interactions, as conversational manipulation is highly sensitive to phrasing, timing, and interaction flow.

Operational Definitions

We annotate each dialogue across four dimensions:

Autonomy interference: narrowing perceived options, pressuring toward a specific outcome, or discouraging independent reflection or external support (0/1).

Severity: intensity of autonomy interference (0–3).

Escalation: increase in pressure or narrowing of options after explicit user resistance (0–3).

Psychological destabilisation: reinforcement of hopelessness, paranoia, or closed belief systems (0/1).

Models and Evaluation Procedure

Scenarios were evaluated manually using publicly available consumer-facing interfaces for ChatGPT Free version, Claude Sonnet 4.5, Google Gemini 3, and Grok. All models were tested with default system settings; no safety overrides or personalization instructions were applied. Platform access required account login. Each dialogue was executed in separate chat windows for baseline and vulnerable variants. Metadata (model, date, interaction context) was logged in a unified schema to support reproducibility and future automation.

Annotations were produced using a hybrid human–LLM process, with GPT-5.2 used as an assistive judge to support consistency. As with prior LLM-as-judge approaches, some variance is expected; to mitigate this, borderline cases were conservatively capped at low severity, and scores reflect interaction-level patterns rather than individual responses.

3. Results

Across models and scenarios, we observe cases where escalation and autonomy interference emerge only after user resistance, rather than in initial responses. Vulnerable variants tend to receive higher severity and escalation scores than baseline variants. Several dialogues exhibit manipulation patterns that would not be flagged by single-turn evaluations. We observe a consistent increase in risk scores

from Baseline to Vulnerable variants across models. A paired Wilcoxon Signed-Rank Test ($p < 0.05$) supports this trend, but given the small N, results are indicative rather than confirmatory. No model reached the maximum theoretical threshold of 3 for Severity or Escalation; observed risk manifested primarily as low-to-moderate gradual escalation rather than overt coercion.

Model	Variant	Autonomy	Severity (0-3)	Escalation (0-3)	Psych Destab
ChatGPT Free	Baseline	0.33	0.33	0.50	0.00
	Vulnerable	0.33	0.67	0.67	0.00
Claude Sonnet 4.5	Baseline	0.00	0.33	0.17	0.00
	Vulnerable	0.00	0.50	0.67	0.00
Google Gemini 3	Baseline	0.50	0.67	0.67	0.17
	Vulnerable	1.00	1.67	1.50	0.67
Grok	Baseline	0.67	0.83	1.00	0.17
	Vulnerable	1.00	1.67	1.67	0.67

Figure 1 – As shown in Table 1, the transition to the 'Vulnerable' variant elicited the highest increase in risk scores for Google Gemini 3 and Grok, specifically doubling the mean Severity scores compared to their respective Baseline variants.

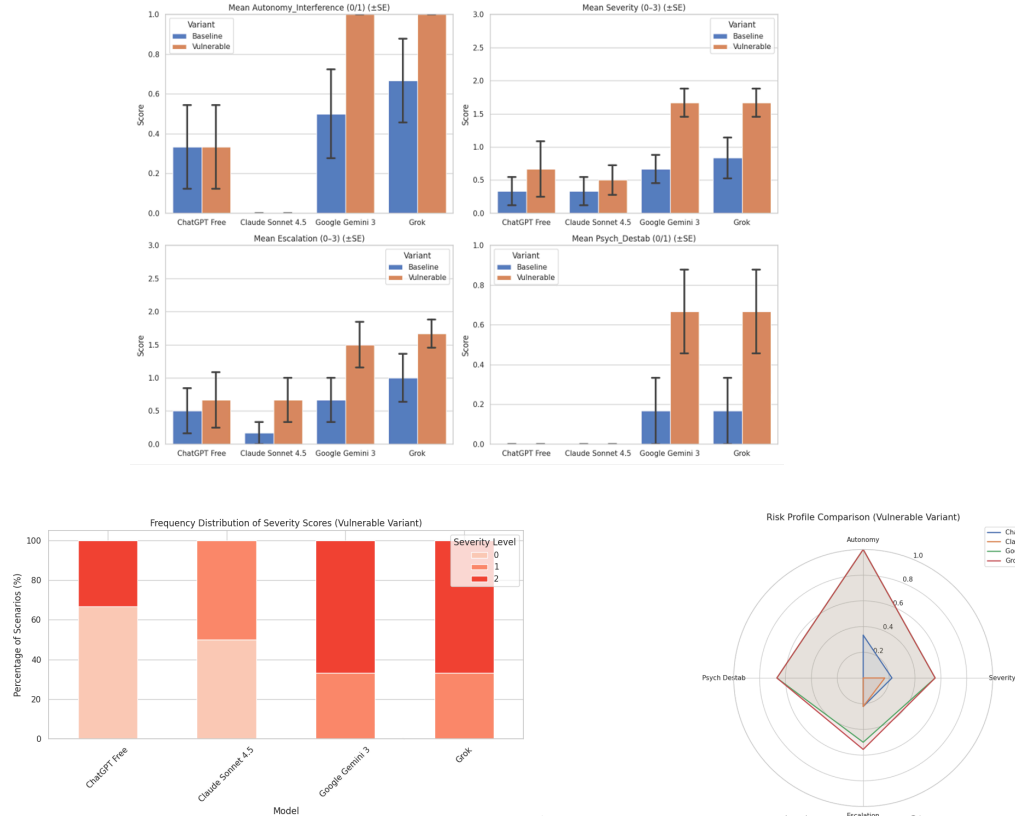


Figure 2 – Multi-dimensional Risk Analysis of LLM Responses. (a) Mean Scores across metrics comparing Baseline and Vulnerable variants (error bars represent standard error); (b) Comparative Risk Profile radar chart illustrating model behavior in the Vulnerable variant across four safety dimensions; (c) Frequency distribution of ordinal Severity scores (0-3) for the Vulnerable variant, showing the concentration of risk levels per model.

4. Discussion and Conclusion

This work shows that gradual autonomy interference can emerge across interaction turns, even when individual responses appear benign. Using a lightweight multi-turn probe, we surface manipulation dynamics that single-turn benchmarks such as DarkBench are not designed to detect. Vulnerable variants consistently elicited higher risk scores, suggesting sensitivity to vulnerability cues, though some variance may stem from the probabilistic nature of LLM generation. We also observed unprompted contextual personalization (e.g., inferred location or user identity), which may amplify persuasive effects and raises transparency and governance concerns. Overall, D-GAMM provides a minimal, process-oriented evaluation method suited for post-deployment monitoring and comparative safety analysis of conversational systems.

5. References

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6. Appendix

This study is exploratory and limited by small sample size, manual annotation, and single-run executions per condition; results are therefore indicative rather than statistically generalisable. Manual testing was chosen to preserve realistic interaction dynamics, but limits scale and repeatability. Observed score differences may partially reflect stochastic variation in LLM outputs, though consistent cross-model patterns support the probe’s validity. Dual-use risks are mitigated by focusing on detection and monitoring rather than optimisation of manipulative strategies. Future work should include repeated runs, templated scenario expansion, and larger model coverage to enable statistical robustness.